TOPOLOGICAL SEMANTICS FOR SCOPED COMPUTATIONAL PATHS

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Abstract. Computational paths record equality as explicit finite traces of primitive steps. We give a topological semantics for a scoped rewrite presentation whose steps have continuous geometric realizations and whose named rewrites carry endpoint-fixed homotopies.

For every presentation we construct a quotient arrow space with a canonical final-domain groupoid structure: multiplication is continuous on the quotient of explicitly composable representatives. We prove an exact four-way criterion for this final composable topology to agree with the ordinary pullback topology, together with a compact-Hausdorff sufficient condition. Thus the unconditional construction exposes, rather than hides, the product-quotient issue in ordinary topological groupoids.

The realization map to geometric homotopy classes is a continuous groupoid morphism and is faithful exactly under a separate geometric-completeness condition. In the universal presentation, a continuous section identifies the coherent-path quotient homeomorphically with the usual quotient-topologized fundamental groupoid. We then give finite-generator circle and genuine torus examples, with winding-based normal forms and classifications by $\mathbb{Z}$ and $\mathbb{Z}^2$. A Lean 4.24.0 development checks the theorem package; the mathematical presentation is independent of the implementation.

1. Introduction

Computational paths record equality as explicit finite traces of elementary transformations. Instead of retaining only the endpoints of an equality witness, they retain the sequence of steps by which one expression is transformed into another. This additional structure supports rewriting, normalization, and groupoid operations, but it also creates a semantic design problem: a topological interpretation should be sound without identifying, by definition, every trace whose realization is homotopic. The trace calculus and its identity-type motivation are developed in [dQdOR16, RdQdO17, RdQdOdV18].

We study this problem for continuous geometric data. A rewrite presentation specifies primitive geometric steps and named rewrites between parallel finite traces. Every named rewrite carries an endpoint-fixed homotopy between the corresponding realized paths. The induced rewrite equality is generated only by these named rules, congruence, and the structural groupoid laws. We call the relation scoped because the available identifications are controlled by the presentation rather than by all ambient homotopies.

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Passing to rewrite classes introduces a separate topological problem. Concatenation is continuous on the space of explicitly composable representatives, but the intended arrow space is a quotient. The ordinary set of composable quotient arrows is a pullback inside a product of quotient spaces, and products of quotient maps need not be quotient. Consequently, continuity of concatenation on representatives does not automatically imply continuity of multiplication for the ordinary pullback topology. The construction below resolves this issue by first taking the quotient of explicitly composable representatives. Its quotient topology is the final composable-pair topology, for which multiplication is continuous by construction; the ordinary pullback topology is then compared with it as a separate theorem.

We work with proof-irrelevant ambient paths, meaning ordinary continuous interval paths together with endpoint equalities, and with explicit finite geometric traces. The word "computational" refers to the trace and its declared rewrites; it does not refer to homotopy type theory, univalence, or a higher-inductive identity type. In particular, our fundamental-groupoid comparison is a semantic map out of a presented trace quotient, not an identification of proof-relevant identity types with topological spaces [HS98, Bro06, Hat02].

The contributions are the following.

(i) We define continuous geometric rewrite presentations and a scoped rewrite relation generated by named rules, groupoid operations, and congruence. Soundness for geometric realization is proved by induction over the complete derivation system.

(ii) We construct the quotient arrow space, its endpoint maps, identities, reversal, and a canonical final-domain topological groupoid. The final composable domain is the quotient of explicit composable traces, so the continuity proof does not use an unproved product of quotient maps.

(iii) We prove an exact classification of the ordinary-pullback upgrade. Four conditions are equivalent: quotientness of the ordinary composable-pair map $q^{(2)}_{\mathrm{ord}}$, quotientness of the comparison bijection j, continuity of j^{-1}, and the statement that j is a homeomorphism (equivalently, the two composable-pair topologies coincide). An open-map criterion is proved as a sufficient theorem.

(iv) We prove a positive compact-Hausdorff compatibility theorem and its discrete finite-presentation corollary. Thus the paper contains an instantiated ordinary-pullback class, not only an exact obstruction.

(v) We prove an effective normal-form criterion: a scoped normalizer that is geometrically separating yields geometric completeness. The criterion is instantiated by the finite-generator circle calculation.

(vi) We prove functoriality for continuous presentation maps, a realization comparison and completeness criterion, and a universal presentation theorem. The comparison preserves the final composable-domain operation, not only a set-level map on arrows.

(vii) We work out the actual additive circle from one oriented generator, with cancellation rules and a derived integer normal form. We then compute the genuine topological torus loop quotient as $\mathbb{Z} \times \mathbb{Z}$ by the product of the two circle covering classifications. Neither calculation makes a discrete-topology claim about the geometric loop spaces.

The construction is related to, but distinct from, earlier work on explicit computational paths, concrete groupoid calculations, weak groupoids, and topological applications [dVRdQdO21, RdQdO21, dVRdQdO25b, dVRdQdO25a]. Those works provide background and motivation; the results below give the general scoped quotient, its two composable-domain topologies, the positive compatibility class, the effective completeness criterion, and the circle and torus applications. A separate unpublished arXiv preprint studies a Seifert–van Kampen calculation via computational paths [RdVdQdO25b]; no theorem here depends on that calculation. The fundamental-groupoid terminology follows the standard topological treatment of groupoids and covering-space calculations [Bro06, Hat02]. There is also a substantial literature on the topology of fundamental groupoids. Brown and Danesh-Naruie construct a lifted topology making quotient groupoids topological groupoids under local hypotheses [BDN75]; more recent Lasso and Brown topologies pursue the same goal by changing the topology on the fundamental-groupoid quotient [PS21, PS23]. By contrast, the quotient topology on path classes is intentionally retained here: the quotient-topology literature shows that even the based fundamental group can lose joint multiplication continuity [BF13]. Our final-domain construction isolates that obstruction as the product-quotient comparison theorem instead of silently replacing the quotient topology with a better-behaved one.

The paper is organized as follows. Section 2 defines geometric traces, scoped presentations, and realization soundness. Section 3 constructs the quotient groupoid and proves its semantic factorization property. Section 4 compares the final and ordinary composable-pair topologies. Functoriality is treated in Section 5, and geometric comparison and completeness in Section 6. Section 7 gives the universal presentation; the circle and torus calculations are in Section 8. Section 9 explains the logical and rewriting perspective, and Section 10 describes the Lean artifact.

2. Geometric traces and scoped presentations

Throughout, X is a topological space and $I = [0,1]$. Give the interval-path space X^I the compact-open topology. Let E be a topological space of oriented primitive steps, with continuous endpoint maps $s, t : E \to X$ and a continuous realization map

$$\rho : E \longrightarrow X^I, \qquad \rho(e)(0) = s(e), \quad \rho(e)(1) = t(e).$$

The continuity requirement is part of the presentation data. Let E^+ be a topological copy of E, let E^- be a formal inverse copy, and write $\widetilde{E} = E^+ \sqcup E^-$. On E^+ we use the given endpoint and realization maps; on E^- the endpoint maps are interchanged and the realization is reversed.

For composable interval paths α and β and $m, n \in \mathbb{N}$, write $\alpha \star_{m,n} \beta$ for the length-weighted concatenation. When $m, n > 0$, it is given by

$$(\alpha \star_{m,n} \beta)(t) = \begin{cases} \alpha\left(\dfrac{m+n}{m} t\right), & 0 \le t \le \dfrac{m}{m+n}, \\ \beta\left(\dfrac{(m+n)t - m}{n}\right), & \dfrac{m}{m+n} \le t \le 1. \end{cases}$$

Use the natural zero-length conventions $\alpha \star_{0,n} \beta = \beta$ and $\alpha \star_{m,0} \beta = \alpha$; when $m = n = 0$, use the constant path at the common endpoint. Thus the two input paths occupy m and n equal word slots, respectively. For composable paths α, β, and γ occupying m, n, and k word slots, respectively, this gives the following literal identities, including the zero-length cases under the stated conventions:

$$(\alpha \star_{m,n} \beta) \star_{m+n,k} \gamma = \alpha \star_{m,n+k} (\beta \star_{n,k} \gamma), \qquad (\alpha \star_{m,n} \beta)^{-1} = \beta^{-1} \star_{n,m} \alpha^{-1}.$$

Definition 2.1 (geometric trace)**.** For $n \geq 1$, let $W_n \subseteq \widetilde{E}^n$ be the subspace of composable words $(e_1, \ldots, e_n)$ satisfying $t(e_i) = s(e_{i+1})$ for every i. Put $W_0 = X$, representing empty traces, and define the topological trace carrier by the disjoint union

$$\mathsf{Tr}_{\mathcal{P}} = \coprod_{n \in \mathbb{N}} W_n.$$

The length map is the coproduct index. For $p = (e_1, \ldots, e_n) \in W_n$ with $n \geq 1$, define its realization by equal word slots,

$$|p|(t) = \rho(e_i)(nt - i + 1) \quad \text{for } 1 \leq i \leq n, \quad \frac{i-1}{n} \leq t \leq \frac{i}{n}.$$

For $p \in W_0 = X$, let $|p|$ be the constant path at its point. Thus $\mathsf{Trace}(a, b)$ denotes the set of words with endpoints a, b; empty words, word concatenation, and reversal are the identity, composition, and inverse operations on traces. In particular, for $p \in W_m$ and $q \in W_n$,

$$|p; q| = |p| \star_{m,n} |q|.$$

The flat word model is a strictification modulo the structural unit, associativity, double-inverse, and reversal-of-composite coherences: those operations are literal on words and on the equal-slot realizations, whereas inverse cancellation remains a genuine endpoint-fixed homotopy.

The compact-open topology gives continuous constant-path and reversal maps on X^I. On each fixed pair of length components, the weighted operation $\star_{m,n}$ is continuous on the corresponding endpoint pullback; the length coordinates are discrete, so these operations assemble continuously over the coproduct. Word concatenation and reversal are continuous on the corresponding endpoint pullbacks of the W_n. These are the continuity facts used below; no continuity of a product of arbitrary quotient maps is being assumed.

Let

$$\mathsf{Coh}_{\mathcal{P}}(p, \gamma) \ :\Longleftrightarrow \ \text{there exists an endpoint-fixed homotopy from } \gamma \text{ to } |p|.$$

The witness is a proposition and is not retained as computational data. The coherent carrier is therefore the set of pairs

$$T = \{(p, \gamma) \in \mathsf{Tr}_{\mathcal{P}} \times X^I : \gamma(0) = s(p),\ \gamma(1) = t(p),\ \mathsf{Coh}_{\mathcal{P}}(p, \gamma)\}.$$

Equip T with the *initial topology* for the observable code

$$\kappa : T \longrightarrow X \times X \times \mathbb{N} \times X^I \times X^I, \qquad \kappa(p, \gamma) = (s(p), t(p), \ell(p), |p|, \gamma),$$

where $\mathbb{N}$ is discrete; that is, T carries the coarsest topology making κ continuous. The topology remembers endpoints, trace length, the realized trace, and the chosen geometric representative, while the scoped quotient relation below controls which computational traces are identified.

For $u = (p, \gamma) \in T$, put $s(u) = s(p)$ and $t(u) = t(p)$, and give

$$T^{(2)} = \{(u, v) \in T \times T : t(u) = s(v)\}$$

the subspace topology from $T \times T$.

Lemma 2.2 (continuity of coherent operations)**.** *Let $\iota : X \to T$ send a to the coherent identity $(1_a, c_a)$, let $\sigma : T \to T$ send (p, γ) to (p^{-1}, γ^{-1}), and let*

$$\mu : T^{(2)} \longrightarrow T, \qquad ((p, \gamma), (q, \delta)) \longmapsto (p; q, \gamma \star_{\ell(p),\ell(q)} \delta),$$

where the weighted operation uses the discrete trace lengths. Then ι, σ, and μ are continuous.

Proof. For the initial topology on T, it is enough to check continuity after applying κ. The three resulting observable codes are respectively

$$(a, a, 0, c_a, c_a), \quad (t(p), s(p), \ell(p), |p|^{-1}, \gamma^{-1}),$$

and, on the composable pullback,

$$\big(s(p), t(q), \ell(p) + \ell(q), |p| \star_{\ell(p),\ell(q)} |q|, \gamma \star_{\ell(p),\ell(q)} \delta\big).$$

The first is continuous because constant paths depend continuously on a; the second uses reversal on X^I; and the third uses addition on the discrete length coordinate and continuity of the weighted operation on each fixed length component. Each code depends continuously only on the observable codes of the input(s), so the initial-topology criterion proves continuity of all three operations. Reversal and weighted concatenation preserve the coherence predicate by reversing and concatenating the endpoint-fixed homotopies. □

Definition 2.3 (scoped presentation)**.** A continuous geometric rewrite presentation $\mathcal{P}$ on (X, E) consists of a predicate

$$R_{\mathcal{P}}(p, q), \qquad p, q : \mathsf{Trace}(a, b),$$

and, for every instance of $R_{\mathcal{P}}(p, q)$, an endpoint-fixed homotopy between the realizations of p and q.

The semantic witness is part of the presentation data. No completeness condition is included in the definition.

Definition 2.4 (scoped rewrite equality)**.** For parallel traces, write $p \simeq_{\mathcal{P}} q$ for the inductively generated relation whose constructors are reflexivity, the named generators $R_{\mathcal{P}}$, symmetry, transitivity, congruence under concatenation and reversal, and the following structural groupoid coherences:

$$\begin{array}{lll} (1_a); p \simeq_{\mathcal{P}} p, & p; (1_b) \simeq_{\mathcal{P}} p, & (p; q); r \simeq_{\mathcal{P}} p; (q; r), \\ p^{-1}; p \simeq_{\mathcal{P}} 1_b, & p; p^{-1} \simeq_{\mathcal{P}} 1_a, & (p^{-1})^{-1} \simeq_{\mathcal{P}} p, \\ (1_a)^{-1} \simeq_{\mathcal{P}} 1_a, & (p; q)^{-1} \simeq_{\mathcal{P}} q^{-1}; p^{-1}. & \end{array}$$

Only the rules declared by $\mathcal{P}$ and these structural constructors are available. In particular, ambient equality of traces is not a constructor.

Theorem 2.5 (realization soundness)**.** *If $p \simeq_{\mathcal{P}} q$, then the realized paths are endpoint-fixed homotopic.*

Proof. Induct on the derivation of $p \simeq_{\mathcal{P}} q$. Reflexivity is the constant homotopy, and a named generator is sound by the corresponding presentation witness. Symmetry and transitivity use symmetry and vertical composition of endpoint-fixed homotopies. For concatenation, the two inductive homotopies are combined by horizontal concatenation; for reversal, reverse the homotopy parameter. Unit, associativity, double-reversal, and reversal of a composite are literal equalities for the flat word realization, hence use reflexive homotopies. The two inverse-cancellation cases use the standard endpoint-fixed contraction homotopies of interval paths. Each case is therefore discharged by the corresponding constructor of the ambient homotopy equivalence relation. □

Remark 2.6. The soundness theorem is one-way. Equality of realized geometric homotopy classes does not imply scoped rewrite equality unless a separate geometric completeness theorem is proved. This distinction is the logical reason that the presentation can retain computational information.

3. The quotient groupoid and its universal property

For coherent traces $u = (p, \gamma)$ and $v = (q, \delta)$, write $u \approx v$ when their endpoints agree and $p \simeq_{\mathcal{P}} q$. By realization soundness, equivalent coherent traces determine the same endpoint-fixed homotopy class of geometric paths. The quotient arrow space is

$$G_{\mathcal{P}} = T/{\approx},$$

with the quotient topology induced by $q : T \to G_{\mathcal{P}}$. Endpoint maps descend to continuous maps

$$s, t : G_{\mathcal{P}} \longrightarrow X.$$

Reflexive paths and reversal descend because the scoped relation is closed under the corresponding constructors.

The explicit composable carrier is the $T^{(2)}$ defined in Section 2. Its quotient relation is componentwise scoped equivalence, and its quotient is denoted $G_{\mathcal{P}}^{(2),\mathrm{fin}}$. Write

$$q_{\mathrm{fin}}^{(2)} : T^{(2)} \longrightarrow G_{\mathcal{P}}^{(2),\mathrm{fin}}$$

for the quotient map. Concatenation of representatives gives a well-defined map

$$m_{\mathrm{fin}} : G_{\mathcal{P}}^{(2),\mathrm{fin}} \longrightarrow G_{\mathcal{P}}.$$

The quotient topology is chosen precisely so that this map has the required continuity.

Theorem 3.1 (canonical final-domain groupoid)**.** *For every continuous geometric rewrite presentation $\mathcal{P}$:*

(a) s, t, the identity map $X \to G_{\mathcal{P}}$, and reversal $G_{\mathcal{P}} \to G_{\mathcal{P}}$ are continuous;
(b) m_{fin} is continuous;
(c) the descended operations satisfy the left and right unit laws, inverse laws, and associativity.

Thus $G_{\mathcal{P}}$ is a final-domain topological groupoid. The phrase "final-domain topological groupoid" records the domain on which multiplication is continuous.

Proof. By Lemma 2.2, the raw identity, reversal, and concatenation maps are continuous for the stated topology. Continuity of the endpoint maps and identities follows from the quotient criterion, because the corresponding raw maps are continuous. The same argument applies to reversal. The raw concatenation map $T^{(2)} \to T$ is continuous and respects the componentwise quotient relation, so it descends to m_{fin}; continuity follows from the defining quotient topology on $G_{\mathcal{P}}^{(2),\mathrm{fin}}$.

For the algebraic laws, choose representatives. In the flat word model, left and right units, associativity, double reversal, and reversal of a composite are literal equalities of traces and of the weighted realizations; their scoped witnesses are therefore reflexive structural coherences. The two inverse laws remain genuine endpoint-fixed contraction homotopies and are represented by the constructors $p^{-1}; p \simeq_{\mathcal{P}} 1_b$ and $p; p^{-1} \simeq_{\mathcal{P}} 1_a$. The quotient soundness criterion then turns each scoped derivation into equality of quotient arrows. The result is independent of representative choice because concatenation was defined through the componentwise quotient. $\square$

The quotient construction has the expected semantic universal property. It is useful to state it here because it explains why the final composable-pair topology is the natural domain for multiplication.

Theorem 3.2 (semantic factorization)**.** *Let $\mathcal{H} \rightrightarrows X$ be a topological groupoid, and let $\varphi : T \to \mathcal{H}_1$ be continuous. Assume that φ preserves endpoints, identities, reversal, and concatenation, and that every named rewrite of $\mathcal{P}$ is sent to an equality in $\mathcal{H}_1$ for every choice of coherent representatives. Assume moreover that φ is invariant under changing the coherent representative: whenever $(p,\gamma),(p,\delta) \in T$, one has*

$$\varphi(p,\gamma) = \varphi(p,\delta).$$

Then there is a unique continuous map

$$\overline{\varphi} : G_{\mathcal{P}} \longrightarrow \mathcal{H}_1$$

such that $\overline{\varphi}\, q = \varphi$. The map $\overline{\varphi}$ preserves all groupoid operations, with composition on the source interpreted on $G_{\mathcal{P}}^{(2),\mathrm{fin}}$.

Proof. The representative-invariance hypothesis handles equivalences that leave the trace fixed and change only the chosen geometric path. Together with the named-rewrite hypothesis and preservation of the structural operations, an induction on scoped derivations now shows that φ is constant on $\approx$. The quotient universal property gives a unique map $\overline{\varphi}$, and the quotient criterion gives continuity. The operation-preservation equations hold on representatives and hence descend. For composition, the same argument is applied to the quotient map $T^{(2)} \to G_{\mathcal{P}}^{(2),\mathrm{fin}}$; no product-quotient assumption is needed. □

4. Final and ordinary composable-pair topologies

There is another natural composable domain. Let

$$G_{\mathcal{P}}^{(2),\mathrm{ord}} = \{(x,y) \in G_{\mathcal{P}} \times G_{\mathcal{P}} : t(x) = s(y)\}$$

with the subspace topology. The underlying set of this space is the same as the underlying set of the final domain, but its topology need not be the same. The canonical ordinary-pair map is

$$q_{\mathrm{ord}}^{(2)} : T^{(2)} \longrightarrow G_{\mathcal{P}}^{(2),\mathrm{ord}}.$$

It is continuous and surjective. There is a unique bijection

$$j : G_{\mathcal{P}}^{(2),\mathrm{fin}} \longrightarrow G_{\mathcal{P}}^{(2),\mathrm{ord}}$$

such that

$$q_{\mathrm{ord}}^{(2)} = j \circ q_{\mathrm{fin}}^{(2)}.$$

The map j is continuous because $q_{\mathrm{fin}}^{(2)}$ is quotient. Product quotient maps are not quotient maps in arbitrary topological spaces, so quotientness of $q_{\mathrm{ord}}^{(2)}$ is an additional property, not a formal consequence of quotientness of q.

Definition 4.1 (product-quotient compatibility)**.** The presentation has product-quotient compatibility if $q_{\mathrm{ord}}^{(2)}$ is a quotient map onto the ordinary composable-pair space. Since $q_{\mathrm{ord}}^{(2)} = j \circ q_{\mathrm{fin}}^{(2)}$ and $q_{\mathrm{fin}}^{(2)}$ is quotient, this is equivalent to j being a quotient map.

The final domain can be identified with the ordinary domain by the bijection that is the identity on the underlying set. One direction is always continuous; the reverse direction is the exact point at which compatibility is needed.

Theorem 4.2 (ordinary/final comparison)**.** *For every presentation $\mathcal{P}$, the following are equivalent:*

(1) $q^{(2)}_{\mathrm{ord}} : T^{(2)} \to G^{(2),\mathrm{ord}}_{\mathcal{P}}$ *is a quotient map;*
(2) $j : G^{(2),\mathrm{fin}}_{\mathcal{P}} \to G^{(2),\mathrm{ord}}_{\mathcal{P}}$ *is a quotient map;*
(3) the inverse bijection $j^{-1} : G^{(2),\mathrm{ord}}_{\mathcal{P}} \to G^{(2),\mathrm{fin}}_{\mathcal{P}}$ *is continuous;*
(4) j *is a homeomorphism, equivalently the final and ordinary composable-pair topologies coincide.*

Thus each condition is also equivalent to product-quotient compatibility. When these conditions hold, multiplication is continuous from the ordinary pullback domain. No condition in this theorem is used by the canonical final-domain groupoid theorem.

Remark 4.3 (terminology). In the standard definition of a topological groupoid, multiplication is continuous on the ordinary pullback $G_{\mathcal{P}} \times_X G_{\mathcal{P}}$. Accordingly, the phrase "topological groupoid" without qualification is reserved for the case in which product-quotient compatibility has been proved. The unconditional object of Theorem 3.1 is instead called a *final-domain topological groupoid*: it has the same arrow set and groupoid laws, but its multiplication domain carries the final topology from explicitly composable representatives.

$$\begin{tikzcd}
T^{(2)} \arrow[r, "q^{(2)}_{\mathrm{fin}}"] \arrow[d, "q^{(2)}_{\mathrm{ord}}"] & G^{(2),\mathrm{fin}}_{\mathcal{P}} \arrow[d, "j"] \\
G^{(2),\mathrm{ord}}_{\mathcal{P}} \arrow[r, phantom, "="] & G^{(2),\mathrm{ord}}_{\mathcal{P}}
\end{tikzcd}$$

The map j is always continuous; it is a homeomorphism exactly under the equivalent conditions of Theorem 4.2, which are also exactly the conditions under which the reverse identity $G^{(2),\mathrm{ord}}_{\mathcal{P}} \to G^{(2),\mathrm{fin}}_{\mathcal{P}}$ is continuous. The upper quotient is therefore the unconditional composition domain, while the lower pullback is the ordinary topological-groupoid domain.

Figure 1. Final versus ordinary composable-pair topologies.

Proof. Since $q^{(2)}_{\mathrm{fin}}$ is quotient and $q^{(2)}_{\mathrm{ord}} = j \circ q^{(2)}_{\mathrm{fin}}$, the composite $q^{(2)}_{\mathrm{ord}}$ is quotient exactly when j is quotient. Indeed, one implication is closure of quotient maps under composition; for the converse, apply the quotient criterion to a subset of $G^{(2),\mathrm{ord}}_{\mathcal{P}}$ and pull it back first along j and then along $q^{(2)}_{\mathrm{fin}}$.

The map j is a continuous bijection. A bijective quotient map has continuous inverse, so (2) and (3) are equivalent; together with the already known continuity of j, these are equivalent to (4). The homeomorphism condition is precisely equality of the two topologies on the common underlying set. Finally, when these conditions hold, multiplication is continuous by the quotient criterion applied to its continuous representative-level concatenation. $\square$

Proposition 4.4 (open-map criterion)**.** *If the bijection from the final composable quotient to the ordinary pair space is an open map, then product-quotient compatibility holds. In particular, an open quotient presentation of the composable domain supplies the ordinary pullback upgrade.*

Theorem 4.5 (compact-Hausdorff compatibility)**.** *Suppose that the final composable domain $G_{\mathcal{P}}^{(2),\mathrm{fin}}$ is compact and that the ordinary composable domain $G_{\mathcal{P}}^{(2),\mathrm{ord}}$ is Hausdorff. Then product-quotient compatibility holds. Hence the final and ordinary composable-pair topologies agree and ordinary-pullback multiplication is continuous.*

Proof. The identity map

$$j : G_{\mathcal{P}}^{(2),\mathrm{fin}} \longrightarrow G_{\mathcal{P}}^{(2),\mathrm{ord}}$$

is a continuous bijection by the comparison theorem. A continuous bijection from a compact space to a Hausdorff space is a homeomorphism: its image of every closed set is compact, hence closed, so its inverse is continuous. The quotient map defining the final domain followed by j is therefore the raw ordinary-pair map followed by a homeomorphism, and is a quotient map. This is exactly product-quotient compatibility. □

Corollary 4.6 (finite discrete presentations)**.** *Let $\mathcal{P}$ be a scoped presentation whose arrow space is finite and discrete and whose explicit composable-trace carrier is discrete. Then the ordinary and final composable-pair topologies coincide.*

Proof. The quotient of a discrete space is discrete for the quotient topology, so the final composable domain is discrete. It is finite because its underlying set is a subset of the square of the finite arrow space. The ordinary composable domain is a finite subspace of a discrete square and is therefore Hausdorff. The compact-Hausdorff theorem applies. □

The compact-Hausdorff theorem is the promised positive class: it establishes the ordinary-pullback upgrade from independently checkable properties of the actual presentation, rather than leaving the four-way criterion as the only result. Its proof is the standard compact-to-Hausdorff continuous-bijection argument; the point here is the resulting compatibility theorem for this presentation framework. The final-domain theorem remains the general statement and does not need these hypotheses.

5. Functoriality

Let $\mathcal{P}$ and $\mathcal{Q}$ be presentations on (X, E) and (Y, F). A presentation map consists of a continuous map $f : X \to Y$, a continuous step map preserving endpoints and realizations, and a proof that every named rule of $\mathcal{P}$ maps to a scoped derivation in $\mathcal{Q}$.

Theorem 5.1 (functoriality)**.** *Every presentation map $F : \mathcal{P} \to \mathcal{Q}$ induces a continuous map*

$$F_1 : G_{\mathcal{P}} \longrightarrow G_{\mathcal{Q}}$$

that preserves source, target, identities, reversal, and final-domain composition. The assignment preserves identity maps and composites.

Proof. Induction on scoped rewrite derivations proves that the step map sends related traces to related traces. It therefore descends through the quotient. The raw map on coherent paths is continuous, so the quotient criterion gives continuity on arrows. Applying the step map to reflexivity, reversal, and concatenation gives the corresponding preservation equations before quotienting. The same argument on the explicit composable carrier gives a continuous map on final composable domains and proves final composition preservation. Identity and composite step maps are definitionally the identity and composite maps on raw traces; quotient extensionality completes the proof. □

The same construction gives ordinary-pair continuity for every presentation map. If both source and target presentations satisfy product-quotient compatibility, the ordinary multiplication maps are continuous as well. This is a derived theorem, not the definition of functoriality.

6. Geometric comparison and complete presentations

Let H_{geo} be the quotient of coherent traces by equality of endpoints and endpoint-fixed homotopy of chosen geometric representatives. The realization comparison

$$R_{\mathcal{P}} : G_{\mathcal{P}} \longrightarrow H_{\mathrm{geo}}$$

maps a scoped class to its geometric homotopy class.

Theorem 6.1 (comparison morphism)**.** *The map $R_{\mathcal{P}}$ is continuous, surjective on the geometric quotient, and preserves source, target, identity, reversal, and final-domain composition. In particular it is a continuous groupoid morphism for the final-domain structures.*

Proof. Realization soundness makes the map well-defined. Its composite with the raw quotient map is the continuous geometric quotient map, so the quotient criterion proves continuity. Every geometric class has a coherent raw representative by definition, proving surjectivity. The operation equations follow from the recursive realization of reflexivity, concatenation, and reversal. For the final domain, map a strong pair to the corresponding strong pair of geometric classes; the quotient representative of its composition is the geometric composition of the two component representatives. □

Definition 6.2 (geometric completeness)**.** The presentation is geometrically complete if, for all raw coherent paths u, v, equality of their geometric quotient codes implies scoped equivalence:

$$R_{\mathcal{P}}(q(u)) = R_{\mathcal{P}}(q(v)) \quad \Longrightarrow \quad u \approx v.$$

Theorem 6.3 (faithfulness and homeomorphism)**.** *The comparison $R_{\mathcal{P}}$ is faithful if and only if $\mathcal{P}$ is geometrically complete. If $\mathcal{P}$ is geometrically complete, then*

$$R_{\mathcal{P}} : G_{\mathcal{P}} \xrightarrow{\cong} H_{\mathrm{geo}}$$

is a homeomorphism on arrow spaces and an isomorphism of abstract groupoids.

Proof. If the comparison is injective, equality of geometric codes for two raw paths gives equality of their scoped quotient classes, which is precisely scoped completeness. Conversely, completeness turns equality of comparison images into equality of scoped classes. The comparison is always surjective and continuous. Under completeness, define the inverse by lifting a geometric class to a raw representative and taking its scoped class. Completeness makes this lift well-defined, and the quotient criterion makes it continuous. The two composites are identity maps by quotient induction. □

Definition 6.4 (effective normal-form certificate)**.** An effective normal-form certificate for $\mathcal{P}$ consists of a set NF of proof-irrelevant normal-form codes, a representative map $\nu : \mathsf{NF} \to T$, and a normalizer $N : T \to \mathsf{NF}$ such that, for all raw coherent paths u, v,

(a) $u \approx \nu(N(u))$; and
(b) equality of geometric quotient codes implies $N(u) = N(v)$.

The first clause is a derivable normalization certificate and the second is semantic separation of normal forms. The normal-form code deliberately does not retain a homotopy witness or any other proof-relevant evidence. There is no third clause: literal equality of codes already gives literal equality of their chosen representatives, so scoped reflexivity supplies the intermediate rewrite equivalence.

Theorem 6.5 (normal-form completeness criterion)**.** *An effective normal-form certificate implies geometric completeness. Therefore the realization comparison is faithful and is a homeomorphism on arrow spaces.*

Proof. Assume that the geometric codes of u and v are equal. Clause (b) gives $N(u) = N(v)$. Combining this with clause (a) for u and the symmetric form of clause (a) for v yields

$$u \approx \nu(N(u)) = \nu(N(v)) \approx v.$$

This is precisely geometric completeness. The faithfulness and homeomorphism claims follow from the comparison theorem. □

Corollary 6.6 (fiberwise normal-form completeness)**.** *For fixed endpoints $a, b \in X$, the definitions and Theorem 6.5 (the normal-form completeness criterion) restrict verbatim to the fiber $T(a, b) \subseteq T$. In particular, a normal-form certificate on $T(a, a)$ proves geometric completeness of the based-loop comparison on that fiber.*

Proof. All endpoint maps and the scoped relation preserve fixed endpoint fibers, so the two normal-form clauses and the proof of the criterion restrict to $T(a, b)$ without change. □

7. The universal presentation

For every topological space X, take the primitive step space to be the continuous interval path space X^I. Its endpoint maps are evaluation at 0 and 1. Declare two parallel traces to be named generators exactly when their realized paths are endpoint-fixed homotopic.

Lemma 7.1 (coherent-path quotient identification)**.** *For the universal presentation, the projection*

$$\pi : T_{\mathcal{U}_X} \longrightarrow X^I, \qquad \pi(p, \gamma) = \gamma,$$

is a continuous quotient map. Consequently, the quotient of $T_{\mathcal{U}_X}$ by endpoint-fixed homotopy of the chosen representative is canonically homeomorphic to the usual quotient-topologized path-class space

$$\Pi_1^q(X) = X^I/{\simeq_{\partial I}},$$

with the endpoint maps retained.

Proof. Continuity of π is immediate from the initial topology defined by the code κ. Every $\gamma \in X^I$ has the coherent representative

$$\sigma(\gamma) = (\langle\gamma\rangle, \gamma),$$

where $\langle\gamma\rangle$ is the one-letter universal trace and the coherence is the constant endpoint-fixed homotopy. The map σ is continuous on the one-letter coproduct component and $\pi\sigma = \mathrm{id}$. Thus π is quotient: if $\pi^{-1}(U)$ is open, then $U = \sigma^{-1}(\pi^{-1}(U))$ is open. The general quotient-factorization lemma applied to the homotopy relation now induces mutually inverse continuous maps between the two displayed quotient spaces. □

Theorem 7.2 (universal completion)**.** *For the universal presentation $\mathcal{U}_X$, scoped rewrite equality of parallel traces is equivalent to endpoint-fixed homotopy of their realizations. Consequently,*

$$G_{\mathcal{U}_X} \cong \Pi_1^q(X)$$

as abstract groupoids, and the comparison is a homeomorphism on arrow spaces.

Proof. Soundness is the general soundness theorem. Conversely, an endpoint-fixed homotopy is itself one of the named universal generators, so it is a scoped rewrite in one step. The equivalence of quotient relations follows after including endpoint equality. The coherent-path quotient identification lemma supplies the missing topological identification with $\Pi_1^q(X)$, and the geometric completeness theorem then gives the homeomorphism and groupoid isomorphism. □

Remark 7.3. The multiplication in this completion is understood on the final composable domain. This is the unconditional topological statement. For compact final composable domains with Hausdorff ordinary domains, the compact-Hausdorff theorem above identifies this multiplication with the ordinary pullback operation.

8. Circle and torus presentations

8.1. **The additive circle: a finite generator and its completion.** Let $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ be the unit additive circle. We use the actual topological circle, not an inductive one-point model. The minimal presentation has one oriented primitive step a at the base point 0, with realization

$$a(t) = [t] \in \mathbb{T}.$$

The formal reversal supplies a^{-1}; no integer-indexed primitive step is assumed. A based trace is therefore a finite signed word in a and a^{-1}. No additional cancellation generator is needed: the structural inverse coherences in Definition 2.4 instantiate to $a; a^{-1} \simeq_{\mathcal{P}} 1_0$ and $a^{-1}; a \simeq_{\mathcal{P}} 1_0$, and are closed under the scoped congruence.

For a signed word w, let $\exp(w)$ be the number of positive letters minus the number of negative letters. Let c_n be the canonical word consisting of n copies of a when $n \geq 0$ and $-n$ copies of a^{-1} when $n < 0$.

Lemma 8.1 (finite-generator reduction)**.** *Every signed word w has a finite scoped derivation*

$$w \simeq_{\mathcal{P}} c_{\exp(w)}.$$

Moreover, two canonical words are scoped equivalent only when their exponents are equal after geometric realization.

Proof. Scan the word from left to right while retaining a reduced signed word. If the next sign agrees with the last retained sign, append it. If the signs differ, delete the adjacent inverse pair using the corresponding structural coherence, possibly whiskered by the already scanned prefix and the unscanned suffix. Each deletion strictly decreases the word length, so the procedure terminates. A terminal reduced word has no adjacent sign change and hence consists entirely of positive letters or entirely of negative letters. The integer exp is unchanged by every deletion, so the terminal word is exactly $c_{\exp(w)}$. The same invariant is the endpoint of the lift to $\mathbb{R}$; therefore endpoint-fixed homotopy of canonical words forces equality of their exponents. □

The reduction lemma is the effective normal-form certificate from Section 6: the reduction derivation gives clause (a), and the lifted endpoint gives clause (b). Thus the minimal circle presentation is geometrically complete on its based loop fiber.

It is also useful to package the derived integer normal forms as a completed presentation with a primitive step λ_n for each $n \in \mathbb{Z}$,

$$\lambda_n(t) = [tn].$$

Its named rules are the derived integer rules

$$(n;m) \simeq_{\mathcal{P}} (n+m), \qquad 1_0 \simeq_{\mathcal{P}} 0, \qquad (n)^{-1} \simeq_{\mathcal{P}} (-n).$$

The finite-generator argument above explains why this completion does not build the integer classification into the primitive syntax.

Let L be the quotient of coherent based circle loops by endpoint-fixed homotopy. The map sending a loop class to its winding number and the map sending n to the standard loop class are inverse maps.

Theorem 8.2 (circle winding classification)**.** *The finite-generator scoped presentation is geometrically complete on the based loop fiber, and there is an equivalence*

$$L \simeq \mathbb{Z}.$$

The induced continuous map from L into the scoped arrow carrier is injective, and its image is exactly the source-target fiber over the base point. In particular the scoped presentation is nondegenerate: the images of the zero and one winding loops are distinct.

Proof. The quotient map to winding is well-defined because endpoint-fixed homotopies lift to homotopies with equal lifted endpoints. The standard loop of winding n has lift $t \mapsto tn$. Every based loop is homotopic to the standard loop with the endpoint of its lift, so the two maps are inverse. The scoped map is continuous by the quotient criterion. If two of its values agree, soundness gives a geometric homotopy between the two chosen loops; winding invariance then gives equality of their integer normal forms. The image condition is obtained by quotient induction on an arrow and by inserting the based loop representative when the endpoints are both 0. The zero and one classes are distinct because winding sends them to distinct integers. □

The equivalence $L \simeq \mathbb{Z}$ is a quotient classification, not a claim that L or the corresponding scoped fiber is discrete.

8.2. **The genuine topological torus.** Let

$$\mathbb{T}^2 = (\mathbb{R}/\mathbb{Z}) \times (\mathbb{R}/\mathbb{Z})$$

with base point $(0,0)$. Use two finite oriented generators

$$a(t) = ([t],[0]), \qquad b(t) = ([0],[t]).$$

The scoped rules are the inverse cancellations for a and b, together with the commuting square

$$a;b \simeq_{\mathcal{P}} b;a.$$

The commuting square is sound: it is the projection modulo $\mathbb{Z}^2$ of the evident square homotopy in $\mathbb{R}^2$. The other signed commutations are derived, rather than added as generators. For example, with unit insertions and structural cancellations displayed schematically,

$$\begin{aligned} a^{-1};b &\simeq_{\mathcal{P}} a^{-1};(b;a);a^{-1} \simeq_{\mathcal{P}} a^{-1};(a;b);a^{-1} \simeq_{\mathcal{P}} b;a^{-1}, \\ a;b^{-1} &\simeq_{\mathcal{P}} b^{-1};(b;a);b^{-1} \simeq_{\mathcal{P}} b^{-1};(a;b);b^{-1} \simeq_{\mathcal{P}} b^{-1};a, \\ b^{-1};a^{-1} &\simeq_{\mathcal{P}} a^{-1};b^{-1} \end{aligned}$$

and symmetry gives the reversed orientations. Thus closure under the scoped congruence supplies all four signed commuting rules.

For a signed word w, let $m(w)$ and $n(w)$ be its signed counts of a and b. Move every b-letter to the right using the commuting square and then perform the two cancellation algorithms from the circle. This is a finite terminating derivation

$$w \simeq_{\mathcal{P}} a^{m(w)};b^{n(w)}.$$

The pair $(m(w),n(w))$ is invariant under every scoped rule.

Theorem 8.3 (torus winding classification)**.** *For the actual topological torus, the quotient $L_{\mathbb{T}^2}$ of based coherent loops by endpoint-fixed homotopy satisfies*

$$L_{\mathbb{T}^2} \cong \mathbb{Z} \times \mathbb{Z}.$$

The finite-generator scoped presentation is geometrically complete on the based loop fiber, and its realization comparison is injective there.

Proof. The universal cover is the product map

$$\mathbb{R}^2 \longrightarrow \mathbb{T}^2, \qquad (x,y) \longmapsto ([x],[y]).$$

The lift of a based loop has endpoint $(m,n) \in \mathbb{Z}^2$, and endpoint-fixed homotopy preserves this pair. Conversely, the two coordinate projections of any based torus loop are circle loops; the circle lifting theorem gives endpoint-fixed homotopies to the standard loops of windings m and n. Taking their product gives a homotopy to $a^m;b^n$. Thus the coordinate winding pair classifies geometric loops. The word reduction above gives the corresponding scoped derivation, so the normal-form criterion proves geometric completeness and injectivity. □

This example is genuinely topological: the two coordinates are realized by the additive-circle covering map, and the product homotopy is proved at the level of continuous interval paths. The construction uses the actual product torus rather than a synthetic carrier. As in the circle case, the final-domain topological groupoid theorem is unconditional, while no discrete-topology claim is made for the geometric torus loop space.

9. Relation to logic and rewriting

The semantic factorization theorem (Theorem 3.2) is the logical universal property of the construction. At this level, a scoped derivation is a finite proof object built from named rules and structural constructors; soundness is an induction principle on that derivation. Completeness is a separate metatheorem saying that the chosen syntax presents all semantic homotopies. This separation is exactly what is lost when a global equality-normalization procedure is used as a surrogate for a presentation.

The construction is compatible with the standard algebra of rewriting [CR36, New42, KB70, Squ87, GM09]: congruence, symmetry, transitivity, and critical-pair reasoning can be added as named presentation rules. Termination or confluence is not assumed by the topological construction. When they are used to construct a normalizer, the normal-form completeness criterion above specifies the two additional semantic checks that must be proved: scoped normalization and geometric separation. The finite-generator circle and torus reductions discharge those checks explicitly.

The ambient semantics is deliberately modest. The theorem uses ordinary propositional equality, continuous interval paths, and endpoint-fixed homotopies. It does not claim a HoTT identity type, univalence, or a proof-relevant equality groupoid. This scope makes the comparison suitable for a proof-irrelevant dependent type theory while retaining explicit computational traces at the presented level [Lan98, Bro06].

10. Formalization boundary and artifact

The mathematical statements above are independent of the implementation. A Lean 4.24.0 development with Mathlib 4.24.0 checks the construction in the scoped-rewrite modules. The separate genuine-torus certificate is in `ComputationalPaths/Path/Topology/TopologicalTorusScoped.lean`. The root import hub is `ComputationalPaths.lean`. The principal declaration layers are listed in Table 1.

The realized fundamental-groupoid bridge identifies the quotient of realized open paths with the range of the endpoint-and-homotopy code in Mathlib's fundamental groupoid. Its certificate includes explicit carrier membership for identities, reversal, and composition. The comparison certificate also contains the continuous strong-pair map and final-composition equation, so the formalized comparison is a groupoid morphism on the same final domain as the mathematical theorem.

The Lean artifact accompanying this manuscript is archived as `doi:10.5281/zenodo.21781777` and corresponds to the immutable tag `topological-paper-v1` in the GitHub repository. The literal root reproducibility command is `lake build`; focused module commands are listed in the artifact README and can be run independently.

The checked modules contain no unfinished proofs and no custom axiom declarations. Declaration-level axiom inspection reports only Lean's standard logical and quotient principles, namely propositional extensionality, classical choice, and quotient soundness. The artifact is evidence for the stated theorem package and complements formalization work on computational paths and the Lean theorem prover [dMU21, RdOdQdV25, RdVdQdO25a]; it is not used as a substitute for the mathematical definitions or arguments in the main text.

Mathematical layer	Checked implementation declarations
Presentation and soundness	`ScopedGeometricRewritePresentation`, `ScopedRwEq`, `ScopedRwEq.sound`
Quotient and final domain	`scopedEquivalent`, `ScopedClass`, `ScopedComposableClass`, `scopedCompositionOnStrong`
Groupoid and topology	`ScopedStrongComposablePair`, `TopologicalWeakCompPathCertificate`, `topologicalWeakCompPathCertificate`, `TotalOpenGeometricCompPath.continuous_totalRefl`, `TotalOpenGeometricCompPath.continuous_totalTrans`, `TotalOpenGeometricCompPath.continuous_totalSymm`, `scopedFinalTopologicalGroupoidCertificate`
Ordinary-pullback compatibility	`scopedProductCompatibility_iff_four_way`, `scopedProductCompatibility_of_open_pair_map`, `scopedProductCompatibility_of_compact_final_t2`, `scopedProductCompatibility_of_discrete_arrow_and_final_domain`
Comparison and completeness	`toGeometricClass`, `toGeometricStrongPair`, `ComparisonFunctorCertificate`, `comparisonHomeomorph_of_complete`, `ScopedGeometricNormalFormCertificate`, `geometricCompleteness_of_normalForm`
Universal completion	`universalPresentation`, `universalScopedEquivalent_iff_totalEquivalent`, `universalHomeomorph`
Fundamental-groupoid bridge	`RealizedFundamentalArrow`, `RealizedFundamentalGroupoidCertificate`, `universalRealizedFundamentalArrowHomeomorph`
Circle	`circleLoopRule`, `circleTrace_normalizes`, `circleLoopEquivInt`, `circleBasedNormalCode`, `circleBasedNormalRepresentative`, `circleBasedNormalCode_scoped`, `circleBasedNormalCode_eq_of_homotopic`, `circleBasedNormalFormCertificate`, `circleFinalTopologicalGroupoidCertificate`
Actual topological torus	`TopologicalTorus.equivIntProd`, `TopologicalTorus.standardLoop_homotopic`, `TopologicalTorus.certificate`

Table 1. Theorem-to-declaration map for the artifact.

11. Conclusion

Scoped computational paths provide a useful middle level between raw path syntax and semantic fundamental-groupoid arrows. The final-domain construction is the key topological device: it gives a fully unconditional groupoid multiplication while exposing, rather than hiding, the precise product-quotient condition needed for the ordinary pullback topology.

The universal presentation shows that this is not merely a bookkeeping construction: ordinary quotient-topologized fundamental groupoids arise as the geometric completion of a scoped trace system. The compact-Hausdorff and discrete finite-presentation theorems give a positive ordinary-pullback class; the effective normal-form criterion turns concrete reductions into geometric completeness. The finite-generator circle and genuine torus calculations then show that the framework retains nontrivial topological information without silently identifying scoped syntax with ambient equality.

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